

Quiz 3

Imagine the chance experiment of rolling a six-sided die three times.

What sample space (with equally likely outcomes) would you use to model this experiment?

We can use the set of ordered triples (i, j, k) where i, j, k are each integers between 1 and 6. For example, the outcomes $(1, 2, 3)$ means that the first roll is a 1, the second roll is a 2, and the third roll is a 3.

How many outcomes are there in your sample space?

Counting using the multiplication rule, we see that there are $(6)(6)(6) = 216$ outcomes in the sample space.

Let N be the random variable that counts the number of 6's rolled in this chance experiment.

Describe the distribution of N . Explain your probability calculations.

The possible values for N are 0, 1, 2, 3.

$P(N = 0)$: There are $(5)(5)(5) = 125$ outcomes in the event $(N = 0)$, namely (i, j, k) where i, j, k are between 1 and 5. Since all outcomes are equally likely, we have $P(N = 0) = 125/216$.

Another way to calculate this probability is as $(5/6)(5/6)(5/6)$. The probability of not getting a six on each of the three rolls is $5/6$, and these events are independent.

$P(N = 1)$: The outcomes in the event $(N = 1)$ are triples (i, j, k) where i, j, k are integers between 1 and 6 with exactly one 6. We can count these outcomes using the multiplication rule:

- First pick one of the three slots for the 6 ($C_{3,1} = 3$ choices).
- Then fill in the other two slots with integers between 1 and 5 ($(5)(5) = 25$ choices).
- By the multiplication rule, there are 75 outcomes in the event $(N = 1)$.

The probability $P(N = 1)$ is thus $75/216$.

We could also calculate $P(N = 1)$ in the style of questions from Homework 5 and Homework 6. We view $(N = 1)$ as the (disjoint) union of the events "roll 6 only on the first roll", "roll 6 only on the second roll", and "roll 6 only on the third roll". Using the independence of the outcomes for individual rolls, we see that each of these events has probability $(1/6)(5/6)(5/6)$, and so $P(N = 1) = (1/6)(5/6)(5/6) + (1/6)(5/6)(5/6) + (1/6)(5/6)(5/6) = 75/216$.

$P(N = 2)$: The outcomes in the event $(N = 2)$ are triples (i, j, k) where i, j, k are integers between 1 and 6 with exactly two 6's. We can count these outcomes using the multiplication rule:

- First pick two of the three slots for the 6's ($C_{3,2} = 3$ choices).

- Then fill in the other slot with an integer between 1 and 5 (5 choices).
- By the multiplication rule, there are 15 outcomes in the event $(N = 2)$.

We could also calculate $P(N = 2)$ using Homework 5/Homework 6 strategies combining the algebra of events (writing $(N = 2)$ as a disjoint union of three simpler events) and independence.

People who have not studied probability theory often guess that the probability of rolling at least one six in three rolls is $1/6 + 1/6 + 1/6 = 1/2$. According to your model, what is the probability of rolling at least one 6? Why can we not simply sum the probability of getting a 6 on each roll? Explain using the vocabulary of Chapter 1 of Durrett (sample space, events, algebra of events, rules of probability, etc.). You may answer on the back!

We can calculate $P(\text{at least one } 6)$ as $P((N = 1) \cup (N = 2) \cup (N = 3))$. Since these events are disjoint this probability is $P(N = 1) + P(N = 2) + P(N = 3)$, which works out to be $91/216$.

Another strategy is to use our “at least one” trick, calculating $P(\text{at least one } 6)$ as $1 - P(N = 0)$, which again works out to be $91/216$.

Probability novices try to calculate this probability as $1/6 + 1/6 + 1/6 = 1/2$. That number is too large. Here is an analysis of their reasoning: let A be the event “first roll is a 6”; let B be the event “second roll is a 6”; and let C be the event “third roll is a 6”. The event “at least one 6” is $A \cup B \cup C$. A novice might guess that $P(A \cup B \cup C)$ is $P(A) + P(B) + P(C) = 1/6 + 1/6 + 1/6$. Having studied the rules of probability, we know that we can only add probabilities to calculate the union of events when the events are *disjoint*. The events A, B, C , however are not disjoint: for example, the outcome $(6, 6, 6)$ is in all three events. Because the events are not disjoint, there are correction terms to calculate the probability of their union. We know the correction term for the union of two events ($P(A \cup B) = P(A) + P(B) - P(A \cap B)$). We will learn about correction terms for calculating the probability of a union of more than two events in Chapter 2.